PO4: Supported through design of scalable cloud solutions.
PO5: Achieved through usage of cloud platforms and management tools.
PO6–PO8: Indirectly addressed through presentations, teamwork and project activities.

Syllabus

Module No		Hours
1	Principles of Parallel and Distributed Computing: Eras of Computing, Parallel vs. Distributed Computing, Elements of Parallel Computing, Elements of Distributed Computing, Technologies for Distributed Computing. Cloud Computing at a Glance, Building Cloud Computing Environments, Computing Platforms and Technologies.	9
2	Fundamental Concepts and Models: Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, The Cloud Provider Perspective, The Cloud Consumer Perspective, Cloud Deployment Models.	9
3	Basic Terms and Concepts: Threat Agents, Cloud Security Threats, Additional Considerations, Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM).	9
4	Network: Broadband Networks and Internet Architecture, Web Technology, Multitenant Technology, Service Technology. Cloud Management Mechanisms: Remote Administration System, Resource Management System	9
5	Architectures: Workload Distribution Architecture, Resource Pooling Architecture, Dynamic / Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture	9

Text Books

1. Cloud Computing Concepts, Technology & Architecture By Thomas Erl, Zaigham
2. Mastering Cloud Computing by Christian Vecchiola, Rajkumar Buyya, and S. Thamarai Selvi,2013,MK. ISBN: 978-0-12-411454-8.

Reference Books (RB):

1. The Basics of Cloud Computing: Understanding the Fundamentals of Cloud Computing in Theory and Practice, Derrick Rountree, Ileana Castrillo, Syngress, 2013.

Web Links:

1. **NPTEL Cloud Computing – IIT Kharagpur (Prof. Soumya Kanti Ghosh)**
   NPTEL Cloud Computing Course

2. **SWAYAM Cloud Computing Course**
   SWAYAM Cloud Computing

3. **Microsoft Learn – Describe Cloud Computing**
   Microsoft Learn Cloud Computing Fundamentals